SOFTWARE REQUIREMENTS SPECIFICATION

Souqly

AI-Powered Small Business Marketplace

Target Market: Egypt

Version	2.5 — Complete
Project Type	Web Application (ASP.NET Core MVC)
Languages	Arabic (RTL) and English (LTR)
Document Status	Complete and implementation-ready

This document defines the functional and non-functional requirements for **Souqly**, a bilingual online marketplace that connects small business owners in Egypt with customers. It covers multi-seller checkout, payments, delivery, commissions, product approval, seller earnings hold and settlement, cancellations, failed shipments, AI assistive tools, and administration.

Version 2.5 consolidates product approval, earnings hold/settlement, full customer cancellation lifecycle, and administrator options when a shipment fails (Retry or Cancel).

1. Project Overview

1.1 Purpose

Souqly is a bilingual online marketplace connecting small business owners across Egypt with customers. Sellers create storefronts, submit products for admin approval, manage orders, and may use AI tools to improve listings, analyze images, suggest prices, and support discovery. Customers buy from multiple sellers in one checkout. The platform handles payments, orders, delivery, commissions, refunds, earnings hold and settlement, notifications, and admin operations.

1.2 Objectives

- Help small businesses establish an online presence.
- Let customers discover products from sellers across Egypt.
- Support multi-seller checkout in a single cart.
- Provide reliable payment, order, and delivery management.
- Use AI to improve listing quality, pricing, and product discovery after the core marketplace is stable.
- Offer Arabic and English interfaces with proper RTL support.
- Give administrators control over seller approval, product approval, policies, earnings settlement, and operations.

1.3 Scope

In scope: General physical products (handmade crafts, accessories, clothing, gifts, and other permitted merchandise).

Out of scope: Food-specific features; post-delivery return shipping / RMA workflow; automated bank payouts to sellers; GPS tracking; live chat between customer and seller; guest checkout. The platform does not manufacture products or own seller inventory. Users must register or sign in to place an order.

2. User Roles

Guest

- Browse public storefronts and products
- Search and filter the catalog
- View product details and seller information
- Must register or sign in to place orders

Customer

- Manage profile and multiple delivery addresses
- Add products from different sellers to one cart
- Place orders (COD or online payment)
- Track orders and deliveries
- Request cancellations or eligible refunds
- Review eligible purchases
- Use the AI shopping assistant after core is stable

Seller

- Submit seller registration application
- Manage approved storefront and products
- Submit products for administrator approval before they go public
- Use AI for listings, image analysis, and pricing (assistive only)
- Define prices, stock, and delivery coverage by governorate
- Accept or reject full seller orders; update processing status
- Respond to customer cancellation requests
- View sales, commissions, refunds, and earnings (Held / Available / Settled)

Administrator

- Approve, reject, suspend, or reactivate sellers
- Approve or reject product listings; hide approved listings
- Configure commission rate, delivery fees, timeouts, hold days, and notification channels
- Manage delivery-provider accounts and assign shipments
- Handle cancellation disputes and failed-shipment decisions (Retry or Cancel)
- Mark seller settlements; view operational and financial reports

Delivery Provider

- View only assigned shipments
- Update delivery statuses: Assigned → Picked Up → In Transit → Delivered / Failed
- Mark COD amount as collected on successful delivery

- Cannot access unrelated orders or seller financial data

External Systems

- Payment gateway for online payments and electronic refunds
- AI provider for listings, image analysis, pricing, and conversational discovery
- Email and SMS providers for notifications

3. Core Business Rules

Defined once here. All functional requirements implement these rules. This section is the single source of truth.

3.1 AI Principles

- AI is assistive only. It never publishes products, never changes prices, never places orders, and never charges customers.
- Seller or customer must explicitly confirm any AI-generated content or price before it takes effect.
- AI suggestions are never presented as verified facts.
- AI failures must not lose user input or corrupt saved data.
- All AI calls use configurable timeouts, usage limits, and error handling.
- All financial calculations use deterministic application logic — never AI.

3.2 Order, Status and Snapshot Rules

- One checkout may contain products from multiple sellers.
- A multi-seller checkout creates one Parent Order and separate Seller Orders.
- Each confirmed order receives a unique human-readable order reference number.
- Order items always store a snapshot of product name, price, and quantity at purchase time. Later product edits do not change history.
- Parent order totals must reconcile with seller-order amounts, delivery fees, and charges.
- Prices, stock, seller status, product Approved status, and delivery eligibility are validated at checkout.
- Repeated requests must not create duplicate orders or charges (idempotency key at checkout).

Status model

Seller Order: Pending → Accepted → Processing → Shipped → Completed, or Cancelled from Pending, Accepted, or Processing as policy allows.

Shipment: Unassigned → Assigned → Picked Up → In Transit → Delivered / Failed.

Parent Order: Derived from seller orders — all Completed = Completed; all Cancelled = Cancelled; mix of active and Completed = Partially Completed; otherwise Open. Recalculated whenever a seller order status changes.

- Status transitions must follow allowed paths; invalid jumps are rejected.
- Seller moves Pending → Accepted → Processing only. Shipment Picked Up sets seller order Shipped. Shipment Delivered sets seller order Completed. Seller cannot set Shipped or Completed manually.
- If a seller does not accept a Pending seller order within the acceptance timeout (default 48 hours, administrator-configurable), the system auto-cancels that seller order.

Seller accept or reject (order acceptance)

- When Pending, the seller can Accept the whole seller order or Reject it with an optional reason.
- Accept moves the seller order to Accepted. Reject cancels it and applies pre-acceptance refund and stock rules.
- Partial line-item accept is out of MVP.

Customer cancellation

Cancellation, refund processing, and stock restoration are separate operations. A Seller Order may be Cancelled while a refund is still pending confirmation from the payment gateway or electronic refund process.

- While the Seller Order is Pending (before seller acceptance): the customer can cancel with one action. Cancellation is automatic.
- While the Seller Order is Accepted or Processing (and not yet Shipped): the customer may submit a Cancellation Request.
- After Shipped or Delivered: no normal customer cancellation flow. Post-delivery issues are manual administrator exceptions only. Full RMA is out of MVP.
- Cancelling one Seller Order does not cancel other Seller Orders under the same Parent Order. Parent Order status is recalculated.

Cancellation request lifecycle

Status	Meaning
Pending	Awaiting seller or administrator decision
Approved	Cancellation accepted; Seller Order becomes Cancelled
Rejected	Request denied; Seller Order keeps its previous status
Withdrawn	Customer withdrew the request while it was still Pending
Escalated	Escalated to administrator (timeout or customer dispute after reject)

- Each cancellation request stores a reason and creation timestamp.
- Only one active cancellation request (Pending or Escalated) is allowed per Seller Order.
- Submitting a cancellation request does not change the Seller Order status. Status changes only when the request is Approved, or on automatic Pending cancel, or on administrator cancel.

• While a cancellation request is Pending or Escalated, the seller must not mark the order as shipped. The system blocks transition to Shipped until the request is resolved.

Seller decision on a cancellation request

• The seller may Approve or Reject a Pending cancellation request for their Seller Order.
• Rejection requires a reason. The customer is notified.
• The seller must respond within the cancellation response timeout (default 48 hours, administrator-configurable). This timeout is separate from the order-acceptance timeout.
• If the seller does not respond within the timeout, the request is Escalated to the administrator. It is not auto-approved or auto-rejected.
• The customer may withdraw a Pending request if the Seller Order has not reached Shipped.
• The customer may escalate a Rejected request to the administrator.

Administrator resolution

• The administrator may Approve (cancel the Seller Order) or Reject (keep current status) an Escalated or disputed request.
• The administrator records a reason. Customer and seller are notified.
• Admin-approved cancellation applies the same inventory and payment rules as seller-approved cancellation, unless the administrator explicitly sets a different restock outcome for an exception case.

Effects when cancellation is Approved

• Seller Order status becomes Cancelled.
• Stock: restore reserved stock when cancellation occurs before shipment. After shipment, stock is restored only if the administrator confirms goods are recovered.
• Online payment: if payment was collected, initiate an eligible refund through the payment gateway. Track refund status separately until the gateway confirms success. Do not mark refund Successful only because a refund request was sent.
• COD: if payment was not collected, no payment refund is required; clear the amount due. If COD was already marked collected, record a refund payable, require administrator-approved electronic refund, and mark Settled only after confirmation.
• Commission is adjusted proportionally for the cancelled or refunded amounts on that Seller Order.
• If an earning record exists, reduce or void it according to refund and cancel rules before Settled.

Failed shipment — administrator options

When a shipment status becomes Failed, the delivery provider records a failure reason. The administrator must choose one option:

Option A — Retry the shipment

• Shipment moves Failed → Assigned (assigned again to a delivery provider).
• Seller Order remains Shipped because the product was already picked up.
• No automatic stock restore and no automatic refund from the retry decision alone.

Option B — Cancel the Seller Order

• Seller Order becomes Cancelled.
• Stock is restored only if the administrator confirms goods are recovered or restock is appropriate.
• Refund rules apply as for an approved cancellation (online gateway refund and/or COD refund payable as applicable).
• The administrator records the cancellation reason.
• Parent Order status is recalculated.

3.3 Product Approval Rules

Products are not public until an administrator approves them.

Status	Meaning
Draft	Seller is editing; not public
PendingApproval	Submitted; waiting for administrator decision
Approved	Public; can appear in search and cart
Rejected	Not public; seller may edit and resubmit
Hidden	Administrator hid an approved listing; not public

• Only Approved products appear in the customer catalog, search, and cart validation.
• Seller submits for approval and cannot self-publish to the public catalog.
• Administrator Approve or Reject (optional reason). Rejected products can be edited and resubmitted.
• Material changes after Approved may return the product to PendingApproval.
• Administrator may Hide an approved listing at any time for policy reasons.
• AI suggestions never grant permission to publish restricted products.

3.4 Financial Rules

Commission formula: Commission basis = eligible product subtotal + seller shipment delivery fee. Commission = basis × platform commission rate.

- One administrator-configurable platform-wide commission rate applies to new eligible orders.
- Commission is calculated and stored per seller order (rate, subtotal, fee, basis, amount).
- A commission-rate change does not alter already confirmed orders.
- Refunds trigger proportional commission adjustments.
- Discounts are out of MVP; discounted subtotal equals normal subtotal.
- Customer delivery fees and delivery-provider costs are tracked separately. Provider cost is for admin reporting only.
- Currency is EGP. Money uses decimal(18,2). Never use floating-point for money. Never use AI for financial calculations.

3.5 Seller Earnings Hold and Settlement

Earnings are calculated when a seller order reaches Completed. Automated bank payouts are out of MVP.

Status	Meaning
Held	Inside hold window after Completed (default 14 days, administrator-configurable)
Available	Hold period ended; eligible for settlement
Settled	Administrator marked as paid offline (optional note or reference)

- Net earnings are based on product subtotal + delivery fee – commission – refunds, with exact amounts stored on the earning record.
- A background job moves Held → Available after the hold period. The job is idempotent and safe to re-run.
- Refunds during Held reduce the eligible amount. Do not settle more than the remaining amount.
- Seller can view own Held, Available, and Settled balances and cannot force a bank withdrawal in MVP.
- Administrator can view all sellers' earnings and mark Available rows as Settled.

3.6 Delivery Fee and COD Collection

- Administrators configure fixed delivery fees by destination governorate.
- Each seller shipment has its own delivery-fee record; the fee is preserved in the confirmed order.
- Checkout is blocked if no valid fee is configured for the destination.
- Seller must serve the destination governorate or the item cannot be ordered.
- Only the Administrator assigns shipments to delivery providers.
- On successful COD delivery, the assigned delivery provider marks the amount as collected. Administrators can correct collection status.

3.7 Cancellation, Refund and Dispute Summary

Detailed lifecycle is defined in Section 3.2. Refund unit in MVP is the Seller Order (or remaining eligible amount). Partial quantity refunds are a future stretch item. Disputes are handled manually by administrators. Returns after delivery (RMA / return shipping workflow) are out of MVP.

- Cancellation approval, refund execution, and stock restoration are separate steps with separate statuses where needed.
- Refund is Successful only after payment-gateway confirmation or authorized electronic-refund confirmation.
- A refund cannot exceed the remaining eligible amount. Duplicate refund processing is prevented.

3.8 Administrator Configuration (MVP)

- Platform-wide commission rate
- Fixed delivery fees by destination governorate
- Seller acceptance timeout (default 48 hours)
- Cancellation request response timeout (default 48 hours)
- Earnings hold days (default 14)
- AI usage limits and timeouts
- Notification channel settings (in-app, email, SMS where supported)

3.9 Notification Matrix (MVP)

- Order placed → Customer, each affected Seller
- Seller order accepted / rejected / auto-cancelled (acceptance timeout) → Customer, Seller
- Cancellation request submitted → Seller and Customer confirmation
- Cancellation approved / rejected / withdrawn / escalated → Customer, Seller; Administrator when escalated
- Administrator dispute decision → Customer, Seller
- Product approved / rejected → Seller
- Shipment assigned → Delivery Provider, Seller
- Shipment status updates including Failed → Customer, Seller
- Failed shipment administrator decision (Retry or Cancel) → Customer, Seller, and Provider as applicable

- COD marked collected → Administrator (and Seller summary as applicable)
- Refund requested / processing / completed / failed → Customer, Seller, Administrator as applicable
- Earning became Available or marked Settled → Seller (optional)

Notification failure must not reverse a successful business transaction. Messages must not expose passwords or card details. In-app notifications are required; email and SMS are used where configured.

3.10 Media, Language, Search and Scale

- Product language: at least one of Arabic or English is required for approval; both are recommended.
- Images: up to 5 per product, maximum 5 MB each, formats JPG, PNG, WebP.
- Search relevance in MVP is text match quality plus recency.
- MVP scale target: hundreds of sellers and thousands of products under academic or demo load.
- Order and financial records are retained for audit (recommended minimum: project duration plus one year).
- Basic rate limits apply on registration, login, search, and AI calls.

4. Functional Requirements

FR-01: Authentication and Account Management

- Secure registration and sign-in for customers and sellers.
- Roles: Customer, Seller, Administrator, Delivery Provider.
- Role-based authorization; users cannot access other users' resources.
- Profile management; customers can save multiple delivery addresses.
- Password recovery and account verification.
- Administrators can suspend and reactivate accounts.
- Guest checkout is not supported.

FR-02: Seller Registration and Storefronts

- Users submit seller applications with business and contact details.
- Administrators approve or reject applications.
- Unapproved sellers cannot submit products for approval or accept orders.
- Approved sellers manage their own storefront (information, products, ratings).
- Sellers define the governorates they serve.
- Administrators can suspend seller operations.

FR-03: Product and Category Management

- Approved sellers create and edit products and submit them for administrator approval (see Section 3.3).
- Product fields include name, description, category, price, images, availability, attributes, and status.
- Stock tracking prevents overselling and negative values.
- Administrators manage categories and subcategories and can approve, reject, or hide listings.
- Required fields and prices are validated before submit for approval.
- Image limits and language rules follow Section 3.10.
- Order items snapshot name, price, and quantity at purchase.

FR-04: Product Discovery and Search

- Browse storefronts and Approved products only.
- Keyword search with filters: category, price range, seller location, delivery availability, attributes.
- Sort by price and relevance; paginated results.
- Product details show seller, price, availability, images, description, and delivery information.
- Exclude Draft, PendingApproval, Rejected, Hidden, and suspended-seller listings.

FR-05: Cart and Multi-Seller Checkout

- Add products from multiple sellers to one cart; update quantities or remove items.
- Validate prices, stock, seller eligibility, product Approved status, and destination coverage before checkout.
- Calculate subtotals, delivery fees, and final amount.
- Create one parent order and separate seller orders with unique order reference numbers.
- Show product subtotal and delivery fee per seller shipment before confirmation.
- Recalculate totals when cart or destination changes; prevent duplicate orders using idempotency.
- Support separate cancellation and refund per seller order.
- Each seller manages only their own seller orders; customer sees parent order and all statuses.
- Seller orders follow the status model; auto-cancel if not accepted within the acceptance timeout.
- For online payment, an order may be AwaitingPayment and expire after 30 minutes if the gateway callback never arrives (stock released).

FR-06: Payment Management

- Support Cash on Delivery (COD) and online payments via a payment gateway.
- Record payment method, amount, transaction reference, and status.
- Confirm online payments only via a trusted gateway response.
- Handle success, failure, pending, and cancelled states; prevent duplicates.
- COD tracks amount due and collection status; provider marks collected on delivery; administrator can correct.
- Never store raw payment-card details.
- MVP records seller earnings only; automated payouts are out of scope.

FR-07: Delivery Management

- Administrators manage delivery-provider accounts.
- Administrator assigns each seller shipment to an eligible delivery provider.

- Providers see only assigned shipments and update statuses per Section 3.2.
- On successful COD delivery, provider marks the amount as collected.
- Customers can track delivery progress.
- Fixed delivery fees by destination governorate.
- Validate that the seller serves the destination before accepting the order.
- Multi-seller orders support separate shipments, assignments, and statuses.
- Record delivery failures and reasons. On Failed, administrator chooses Retry (Option A) or Cancel (Option B) per Section 3.2.
- GPS tracking and automatic allocation are future enhancements.

FR-08: Reviews and Ratings

- Customers can review eligible products only when that seller order is Completed (shipment Delivered).
- One review per product per order; linked to author, product, and order.
- Reviews are displayed on product and storefront pages.
- Administrators can hide inappropriate reviews.

FR-09 to FR-12: AI Features

All AI features follow Section 3.1. They are part of MVP scope but are implemented after the core marketplace (authentication, catalog, orders, payments, and delivery) is stable.

- Listing Assistant: basic info → suggested title, description, and tags (Arabic and English) → seller confirms before submit for approval.
- Image Analysis: suggestions for attributes, styles, and colors — suggestions only.
- Pricing Assistance: optional costs → market range or guidance → seller sets the final price. Cost and margin math is application logic.
- Shopping Assistant: natural language → real catalog products with constraints; cannot place orders or change data.
- Shared requirements: validation, timeouts, usage limits, and failure handling.

FR-13: Notifications and Reporting

- Notifications for order, payment, delivery, cancellation, refund, product approval, failed-shipment decisions, and settlement events (see Section 3.9).
- Channels: in-app required; email and SMS where configured.
- Basic administrator operational and financial summaries; seller sales, commission, and earnings views.
- Advanced analytics are future enhancements.

FR-14 and FR-15: Commission and Delivery Fees

Fully defined in Sections 3.4 and 3.6. The system implements those rules.

FR-16: Cancellations and Refunds

- Implement the full cancellation lifecycle in Section 3.2 (automatic Pending cancel; request flow; timeouts; escalation; withdraw).
- Enforce one active cancellation request per Seller Order and the request status model.
- On Approved cancel: set Seller Order to Cancelled; run stock and refund operations as separate steps.
- Track refund states independently until confirmed by the gateway or authorized electronic process.
- Support administrator Option A and Option B when a shipment is Failed.
- Preserve original transaction records and maintain an auditable history of adjustments.

FR-17: Marketplace Policy and Moderation

- Administrators manage marketplace rules and prohibited-product policies.
- System prevents publication or sale of prohibited products.
- Administrators can hide listings or suspend accounts.
- AI suggestions never override marketplace policies.
- There are no food-specific requirements.

FR-18: Seller Earnings and Settlement

Fully defined in Section 3.5. Sellers view balances; administrator marks Settled; no automated bank payout in MVP.

5. Non-Functional Requirements

ID	Requirement
NFR-01 Security	Authentication, authorization, secure sessions, input validation, protection against common web vulnerabilities, and rate limiting on registration, login, search, and AI calls.
NFR-02 Secrets	Passwords securely hashed. API keys, connection strings, and payment credentials stored outside source code.
NFR-03 Privacy	Personal data limited to authorized users and processes. Order and financial records retained for audit.
NFR-04 Performance	Pagination on list pages; indexes on search columns; acceptable behavior with about 1,000 seeded products for demonstration.
NFR-04b Usability	Clear localized Arabic and English error and success messages for validation, payment failure, stock shortage, unauthorized actions, and cancellation or refund outcomes.
NFR-05 Reliability	Failures in AI, email, SMS, or payment integrations must not corrupt unrelated system data.
NFR-06 Data Integrity	Database constraints and transactions protect orders, payments, inventory, commissions, refunds, and earnings.
NFR-07 Bilingual	Full Arabic RTL and English LTR interface support.
NFR-08 Responsive	Usable layouts on desktop, tablet, and mobile.
NFR-09 Maintainability	Modular architecture with clear separation of Presentation, Business Logic, and Data Access.
NFR-10 Testing	Automated tests cover authorization, order totals, commission, stock, payments, cancellation lifecycle, refund confirmation, product approval, earnings hold, and failed-shipment options.
NFR-11 AI Validation	Validate AI responses before use; invalid responses must not corrupt saved data.
NFR-12 AI Control	Configurable timeouts, error handling, and usage limits for AI calls.
NFR-13 Accessibility	Clear labels, understandable errors, keyboard navigation, and sufficient contrast.
NFR-14 Backup	Defined database backup and restoration procedure.
NFR-15 Financial Accuracy	Deterministic logic and suitable decimal precision; never AI-generated money calculations.
NFR-16 Auditability	Important administrator actions and financial adjustments recorded with timestamps and actor information.
NFR-17 Idempotency	Order confirmation, payment processing, and refunds protected against duplicate execution.
NFR-18 Notification Resilience	Failed notifications logged and retried where appropriate without repeating the business transaction.
NFR-19 Secure Notifications	Notifications must not expose passwords, card details, or unnecessary sensitive data.
NFR-20 Financial Consistency	Order totals, commission, delivery fees, refunds, and seller earnings reconcile according to Core Rules.

6. MVP Scope and Future Enhancements

6.1 MVP — Required for Initial Release

Core marketplace

- Authentication and role-based access
- Seller applications and administrator approval
- Storefronts, products, categories, and images
- Product administrator approval workflow
- Search, filtering, and pagination
- Arabic and English interface; responsive UI

Ordering and operations

- Multi-seller cart and checkout; parent and seller orders
- Stock validation and order history
- COD and online payment (test or sandbox during development)
- Delivery-provider accounts; administrator shipment assignment
- Fixed delivery fees by governorate; tracking and status management
- Failed shipment handling with Retry or Cancel
- Full cancellation request lifecycle and refund tracking
- Product reviews when the seller order is Completed
- In-app notifications; email and SMS as configured

Financial management

- Platform-wide commission rate and formula including delivery fee
- Seller earnings with Held → Available → Settled lifecycle
- Administrator settlement marking; delivery-fee tracking
- Refund and cancellation accounting; basic administrator reports

AI capabilities

Part of MVP but implemented after core is stable: listing assistant, image analysis, pricing guidance, conversational discovery, with validation and limits.

Quality and delivery

- Automated tests for critical business rules
- Security and authorization checks
- Logging, backups, and deployment documentation
- Working demonstration using test payment credentials

6.2 Future / Stretch (not required for MVP acceptance)

- Partial quantity refunds within a seller order
- Custom order or special-request form (customer to seller)
- Back-in-stock notifications; gift note; seller about card
- AI background removal, visual search, voice listing
- Automatic delivery assignment and GPS tracking
- Automated marketplace split payments and bank payouts
- Advanced analytics, dispute automation, and full RMA returns

7. Acceptance Criteria

The initial release is functionally acceptable when these scenarios pass:

- 1. Seller cannot make products public before administrator product approval.
- 2. Approved seller can manage products without accessing another seller's listings.
- 3. Administrator can approve or reject a product; only Approved products appear in catalog, search, and cart.
- 4. Customer can purchase from multiple sellers in one checkout (parent order and seller orders).
- 5. Order total reconciles with product subtotals, delivery fees, and charges.
- 6. Commission uses the configured rate and Core Rules formula and is stored on the seller order.
- 7. Each shipment receives the correct destination fee; Administrator assigns the delivery provider.
- 8. COD and online test payment record correct states; provider can mark COD collected.
- 9. Customer one-click cancels a Pending seller order; stock is restored where policy requires.
- 10. Customer submits one cancellation request on an Accepted order; a second active request is rejected.
- 11. Seller rejects a cancellation request with a reason; customer is notified; order status is unchanged.
- 12. Seller does not respond within the cancellation response timeout; request is Escalated to the administrator.
- 13. Customer withdraws a Pending cancellation request before Shipped.
- 14. Administrator approves an escalated cancellation; Seller Order becomes Cancelled; online refund is initiated and tracked until confirmed.
- 15. COD cancellation before collection requires no payment refund; stock is restored if before shipment.
- 16. Shipment Failed → Administrator Option A: Failed becomes Assigned; Seller Order stays Shipped.
- 17. Shipment Failed → Administrator Option B: Seller Order Cancelled; reason stored; stock and refund follow Section 3.2.
- 18. Cancelling one Seller Order leaves other Seller Orders intact; Parent Order status is recalculated.
- 19. Customers track shipments; providers update only assigned shipments.
- 20. Relevant events trigger configured notifications.
- 21. Seller earnings move Held → Available after hold days; administrator can mark Settled.
- 22. AI features, when enabled, remain assistive-only with explicit confirmation.
- 23. Roles cannot access unauthorized resources.
- 24. Duplicate requests, payment failures, invalid AI responses, and notification failures do not corrupt business records.
- 25. Automated tests pass for totals, commission, stock, authorization, cancellation lifecycle, refund confirmation, product approval, and earnings.
- 26. Application supports Arabic RTL and English LTR layouts.
- 27. Product image uploads enforce count, size, and format limits.

8. Final Implementation Assumptions

- 1. All financial, status, cancellation, product-approval, failed-shipment, and earnings rules follow Section 3.
- 2. Guest checkout is not supported.
- 3. Discounts are out of MVP.
- 4. At least one product language is required; both are recommended.
- 5. Images: up to 5 per product, maximum 5 MB, JPG/PNG/WebP.
- 6. Stock decreases at order placement inside a database transaction; restored according to cancellation stage and policy.
- 7. Seller acceptance timeout default is 48 hours (configurable).
- 8. Cancellation request response timeout default is 48 hours (configurable) and is separate from the acceptance timeout.
- 9. While a cancellation request is Pending or Escalated, the system blocks moving that Seller Order to Shipped.
- 10. Seller accepts or rejects the full seller order only.
- 11. Customer one-click cancel is available only while the seller order is Pending.
- 12. Returns after delivery (RMA) are out of MVP.
- 13. Partial quantity refunds are stretch scope.
- 14. COD collection is marked by the delivery provider; administrator can correct.
- 15. Disputes are handled manually by administrators.
- 16. Administrator configures commission rate, delivery fees, acceptance timeout, cancellation response timeout, hold days, AI limits, and notification channels.
- 17. AI features are implemented after the core marketplace is stable.
- 18. MVP records earnings and administrator settlement only — no automated bank payouts.
- 19. Provider cost is for reporting only.
- 20. Confirmed orders preserve original terms; adjustments are recorded separately.
- 21. General merchandise only; no food-specific features.
- 22. Search relevance is text match plus recency.
- 23. Currency is EGP; money uses decimal(18,2).
- 24. Important status changes trigger configured notifications.

- 25. On Failed shipment, administrator must choose Retry (Option A) or Cancel (Option B).

— End of Software Requirements Specification —
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